

Network Design and Configuration Report for SAFETech Ltd

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1 Introduction

1.1 Purpose

The purpose of this project is to develop a network that meets the company's requirements efficiently while maintaining scalability for future expansion at a reasonable cost.

1.2 Overview of Requirements

We are required to design a network that provides reliable communication across multiple floors, with each floor hosting different departments. The network is going to have 150+ devices, with potential for future expansion. The network must be secure, with proper IP addressing and routing protocols configured to ensure seamless communication between devices.

2 Network Design

2.1 Topology Overview

The proposed network topology is a hybrid of **star** and **hierarchical** models, with a stronger emphasis on hierarchical design.

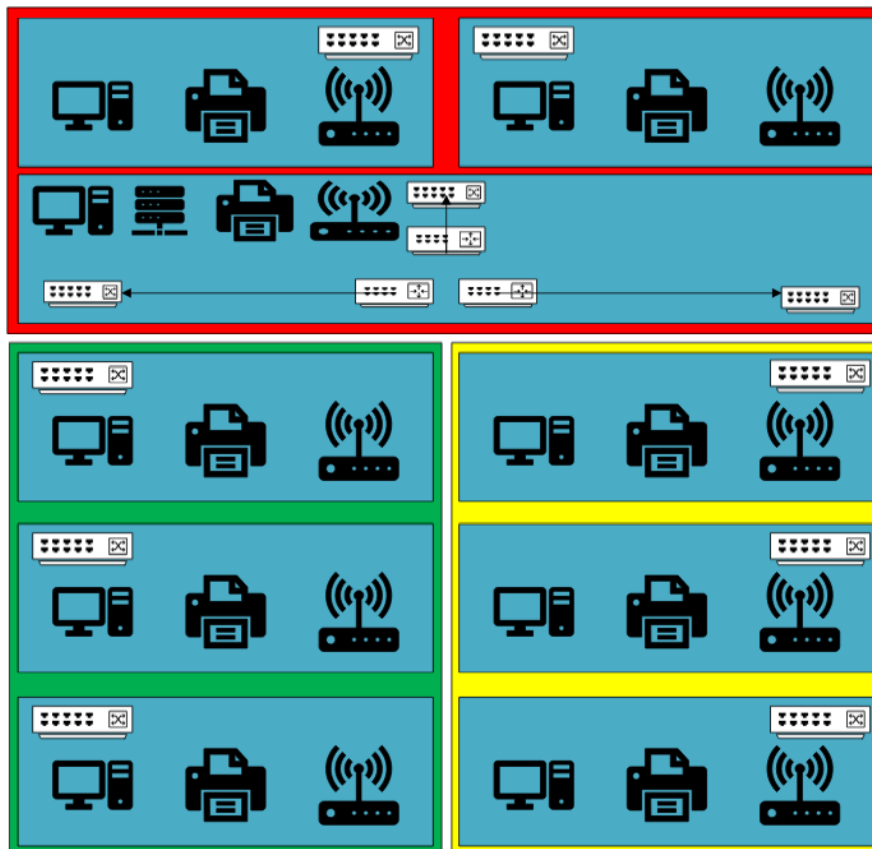


Figure 1: Network Design

Each floor consists of **three subnets**, each connected to dedicated switches. These switches are connected to a **main floor switch**, which is then linked to a router that corresponds to the same floor. This setup has its pros and cons. It is more expensive than simply connecting a single switch to each router and daisy-chaining additional switches as needed. However, daisy-chaining can create a messy and complex cabling setup as more switches are added.

While my approach costs more and is less efficient in terms of switch port usage, it allows for easier expansion. If the company needs to double the number of devices in a department or add new departments to a floor, the structured design will save the tech team significant time. It provides better reliability, ensuring that an issue in one department does not affect others. All of that is outweighing the initial expense and providing a great return on investment.

Floors are color-coded for clarity:

- **Green:** First floor
- **Yellow:** Second floor
- **Red:** Third floor

The final design in Packet Tracer is shown below:

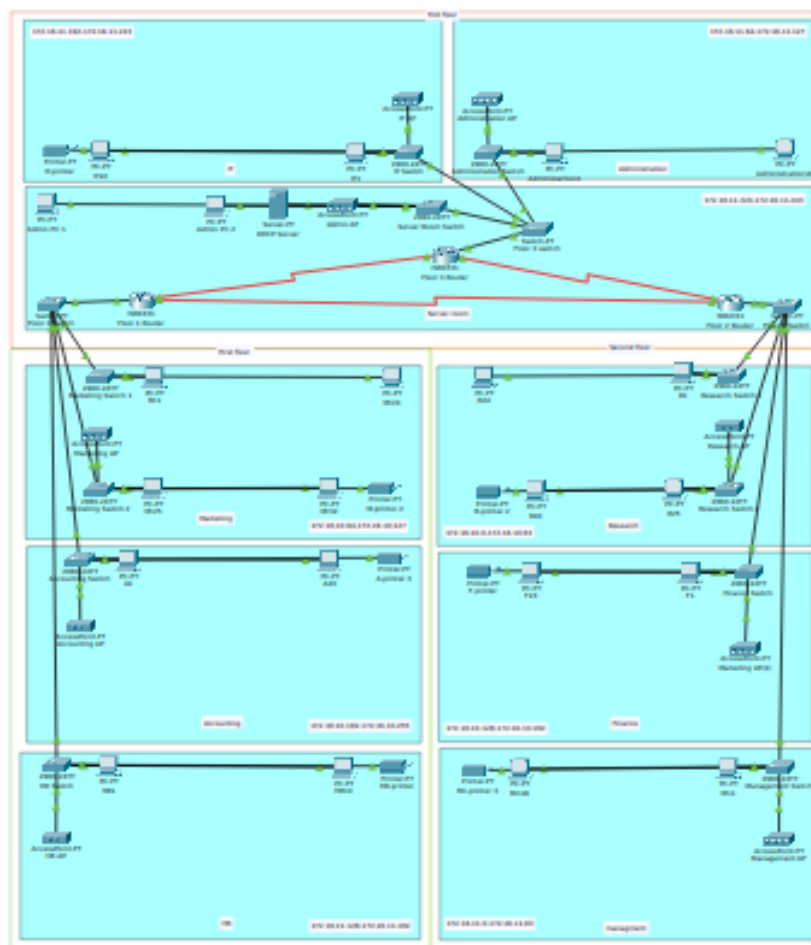


Figure 2: Final Network Design in Packet Tracer

3 Subnetting and IP Addressing

3.1 Subnet Plan

Department	IP Range	Subnet Mask	Devices	VLAN
Research	172.16.10.0 - 172.16.10.63	255.255.255.192	42	21
Marketing	172.16.10.64 - 172.16.10.127	255.255.255.192	34	11
Finance	172.16.10.128 - 172.16.10.191	255.255.255.192	24	22
Accounting	172.16.10.192 - 172.16.10.255	255.255.255.192	23	12
Management	172.16.11.0 - 172.16.11.63	255.255.255.192	21	23
Administration	172.16.11.64 - 172.16.11.127	255.255.255.192	18	33
HR	172.16.11.128 - 172.16.11.191	255.255.255.192	11	32
IT	172.16.11.192 - 172.16.11.223	255.255.255.224	11	13
Server	172.16.11.224 - 172.16.11.243	255.255.255.240	3	31
Routers 1-2	172.16.11.244 - 172.16.11.247	255.255.255.252	2	-
Routers 2-3	172.16.11.248 - 172.16.11.251	255.255.255.252	2	-
Routers 3-1	172.16.11.252 - 172.16.11.255	255.255.255.252	2	-

4 Network Configuration

4.1 Routing Protocol Configuration (OSPF) and VLAN Configuration

The first step is configuring OSPF for inter-router communication:

```
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se0/2/0
Router(config-if)#ip address 172.16.11.250 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/2/0, changed state to up

Router(config-if)#
```

Figure 3: Configuring IP Addresses for Router Communication

```

Router(config)#router ospf 1
Router(config-router)#router-id 2.2.2.2
Router(config-router)#network 172.16.11.248 0.0.0.3 area 0
Router(config-router)#network 172.16.11.244 0.0.0.3 area 0
Router(config-router)#network 172.16.10.0 0.0.0.63 area 0
Router(config-router)#network 172.16.10.128 0.0.0.63 area 0
Router(config-router)#network 172.16.11.0 0.0.0.63 area 0
Router(config-router)#exit
Router(config)#

```

Figure 4: OSPF Configuration on Router 2

VLAN naming follows a **structured convention**:

- The **first digit** represents the floor.
- The **second digit** represents the department.

To ensure smooth inter-VLAN routing, **802.1Q encapsulation** is configured on trunk ports:

```

Router(config)#int g0/0/0
Router(config-if)#no shutdown
Router(config-if)#int g0/0/0.21
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.21, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.21, changed state to up

Router(config-subif)#encapsulation dot1Q 21
Router(config-subif)#ip address 172.16.10.1 255.255.255.192
Router(config-subif)#int g0/0/0.22
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.22, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.22, changed state to up

Router(config-subif)#encapsulation dot1Q 22
Router(config-subif)#ip address 172.16.10.129 255.255.255.192
Router(config-subif)#int g0/0/0.23
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0.23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.23, changed state to up

Router(config-subif)#encapsulation dot1Q 23
Router(config-subif)#ip address 172.16.11.1 255.255.255.192
Router(config-subif)#exit

```

Figure 5: DotQ1 Encapsulation on the Second Floor Router

VLANs are created and assigned names. The trunk port is also configured:

```

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 21
Switch(config-vlan)#name VLAN_21_Research
Switch(config-vlan)#vlan 22
Switch(config-vlan)#name VLAN_22_Finance
Switch(config-vlan)#vlan 23
Switch(config-vlan)#name VLAN_23_Management
Switch(config-vlan)#int gig9/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet9/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet9/1, changed state to up

Switch(config-if)#switchport trunk allowed vlan
% Incomplete command.
Switch(config-if)#switchport trunk allowed vlan 21,22,23

```

Figure 6: VLAN Naming and Trunk Port Setup

VLANs on **secondary switches** are then configured to communicate with the **main switch**:

```

Switch(config)#int g5/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 21
Switch(config-if)#int g6/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 21
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet6/1 (21), with Switch
GigabitEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet5/1 (21), with Switch
GigabitEthernet0/1 (1).

Switch(config-if)#int g7/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 22
Switch(config-if)#int g8/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 23
Switch(config-if)#

```

Figure 7: VLAN Setup on the Main Switch

```

Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range g0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 21
% Access VLAN does not exist. Creating vlan 21
Switch(config-if-range)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 21
Switch(config-if-range)#vlan 21
Switch(config-vlan)#name VLAN_21_Research
Switch(config-vlan)#

```

Figure 8: VLAN Setup on Secondary Switches

4.2 DHCP Helper Address Configuration

To allow VLANs to receive IP addresses from a **centralized DHCP server**, we configure the **IP Helper Address**:

```

Router(config)#int g0/0/0.21
Router(config-subif)#ip helper-address 172.16.11.226
Router(config-subif)#int g0/0/0.22
Router(config-subif)#ip helper-address 172.16.11.226
Router(config-subif)#int g0/0/0.23
Router(config-subif)#ip helper-address 172.16.11.226
Router(config-subif)#exit
Router(config)#ip routing

```

Figure 9: Configuring IP Helper for DHCP Requests

4.3 Basic Configuration for Devices

The first step is setting up the hostname for switches and routers:

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Research_Switch_1
_
```

Figure 10: Setting up Hostname

To set up switches and routers, I used the following set of commands. Packet Tracer has a great feature where you can copy a set of commands, with each new command starting on a new line, which speeds up the process immensely.


```

enable
configure terminal
no ip domain-lookup
banner motd #
+++ ACCESS RESTRICTED +++
Only the faithful may proceed.
Unauthorised access will be purged with righteous fire.
BOW BEFORE THE MACHINE GOD! The flesh is weak;
the Omnissiah's will is eternal.
+++ GLORY TO THE OMNISSIAH +++
#
line console 0
password Zero12345!
login
exit
enable secret ZeroOne2345!
exit
exit
Zero12345!

enable
ZeroOne2345!

conf t
service password-encryption
— only for router —
ip domain-name safetech.com
crypto key generate rsa
1024
username admin privilege 15 secret StrongPassword123
line vty 0 4
transport input ssh
login local
exit
ip ssh version 2
— only for router —
exit
copy run start

reload

```

This set of commands disables domain lookup, creates a banner, sets login and enable passwords, and encrypts all the passwords in the memory of the device. Routers have additional setup steps where SSH is enabled for remote access.

The last step of setup is to save the memory of the device and reload it to ensure correct operation.

5 Server Configuration

5.1 HTTP, DHCP, and Email Server Configuration

The **HTTP** server operates out of the box as soon as an IP address is assigned:

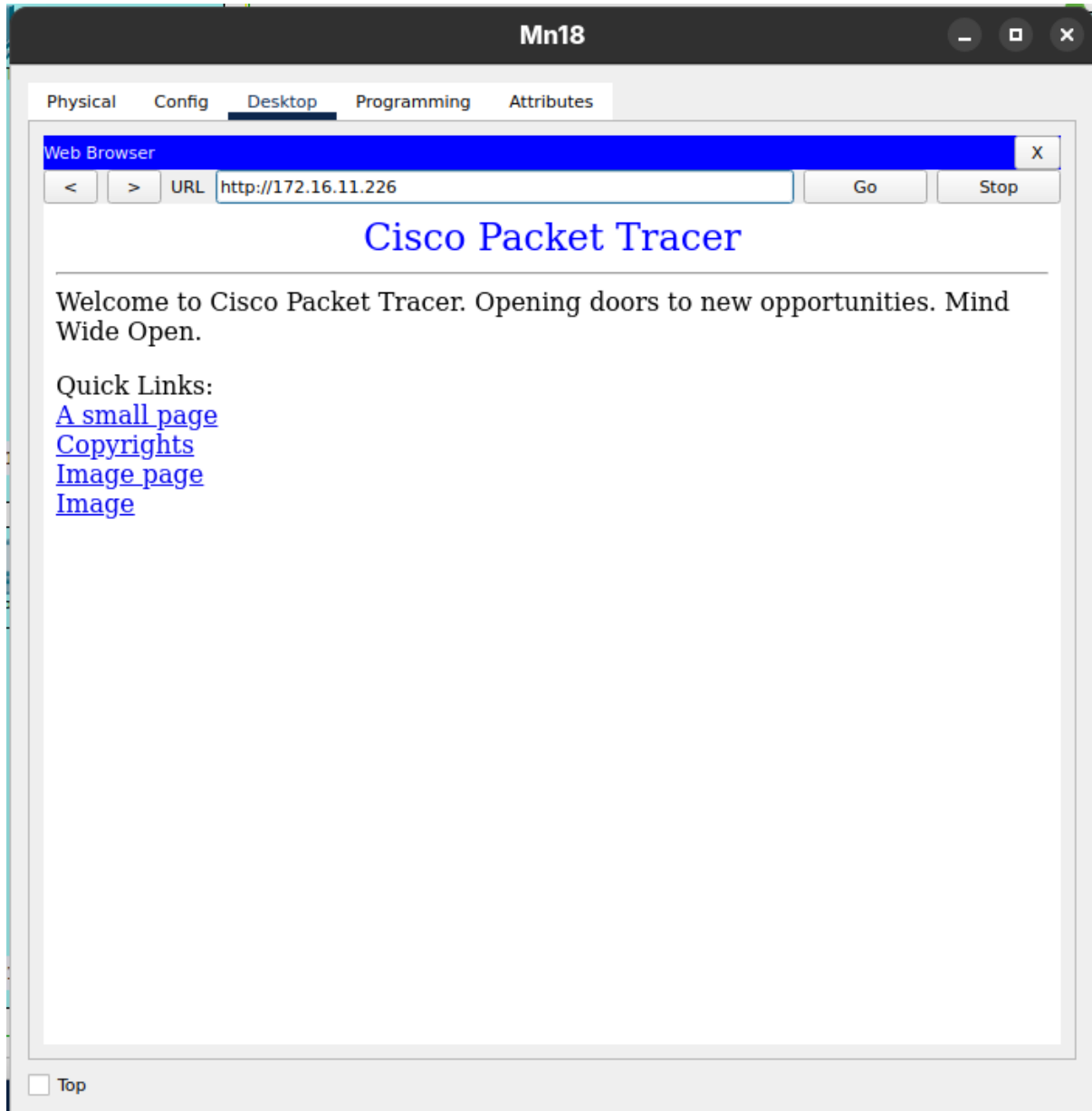


Figure 11: HTTP Server Configuration

The **DHCP** server requires a **static IP address**. We assign 172.16.11.226 as its IP. Each subnet is configured with:

- Subnet mask
- Starting IP
- Number of hosts

- Default gateway

For DHCP to work, we need to add pools of IP addresses in the following way:

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: Floor 2 Research vlan 21

Default Gateway: 172.16.10.1

DNS Server: 0.0.0.0

Start IP Address : 172 16 10 2

Subnet Mask: 255 255 255 192

Maximum Number of Users : 62

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Floor 2 Research via...	172.16....	0.0.0.0	172.16....	255.25...	62	0.0.0.0	0.0.0.0

Figure 12: DHCP Pool Configuration

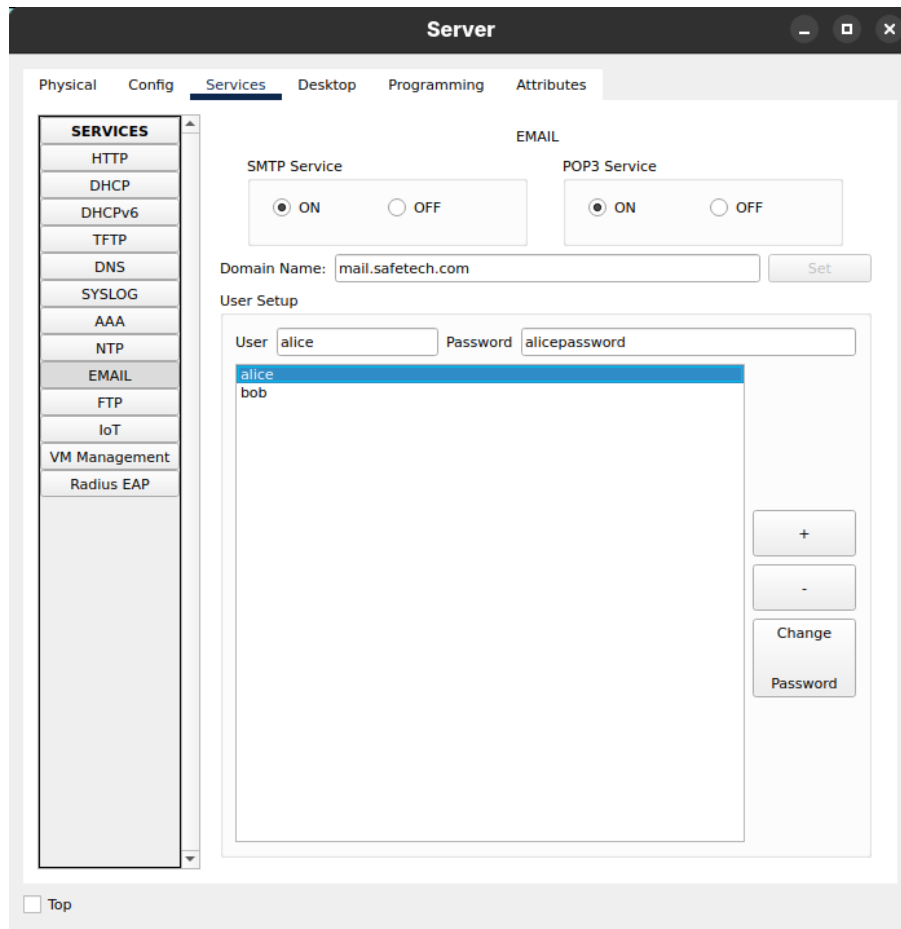


Figure 13: Setting up Email

For the email server to work, we need to enable POP3 and SMTP protocols. The next step is to set the domain to `mail.safetech.com` and add users.

The screenshot shows a window titled "Admin PC 2" with a dark header bar containing standard window controls. Below the header is a tabbed interface with four tabs: "Physical", "Config", "Desktop" (which is selected and highlighted), "Programming", and "Attributes". The "Configure Mail" dialog box is open, featuring a blue title bar with a close button (X). The dialog is organized into three sections: "User Information" with fields for "Your Name:" (containing "alice") and "Email Address" (containing "alice@safetech.com"); "Server Information" with fields for "Incoming Mail Server" and "Outgoing Mail Server" (both containing "172.16.11.226"); and "Logon Information" with fields for "User Name:" (containing "alice") and "Password:" (containing masked characters). At the bottom of the dialog are four buttons: "Save", "Remove", "Clear", and "Reset".

Figure 14: Logging into Email Account

The screenshot shows a window titled "Admin PC 1" with a dark header bar containing standard window controls. Below the header is a tabbed interface with four tabs: "Physical", "Config", "Desktop" (which is selected and highlighted), "Programming", and "Attributes". The "Compose Mail" dialog box is open, featuring a blue title bar with a close button (X). The dialog includes a "Send" button on the left. To the right of the button are two fields: "To:" (containing "alice@mail.safetech.com") and "Subject:" (containing "test"). Below these fields is a large text area containing the text "dighoghdfaoghaq".

Figure 15: Sending Email to Alice

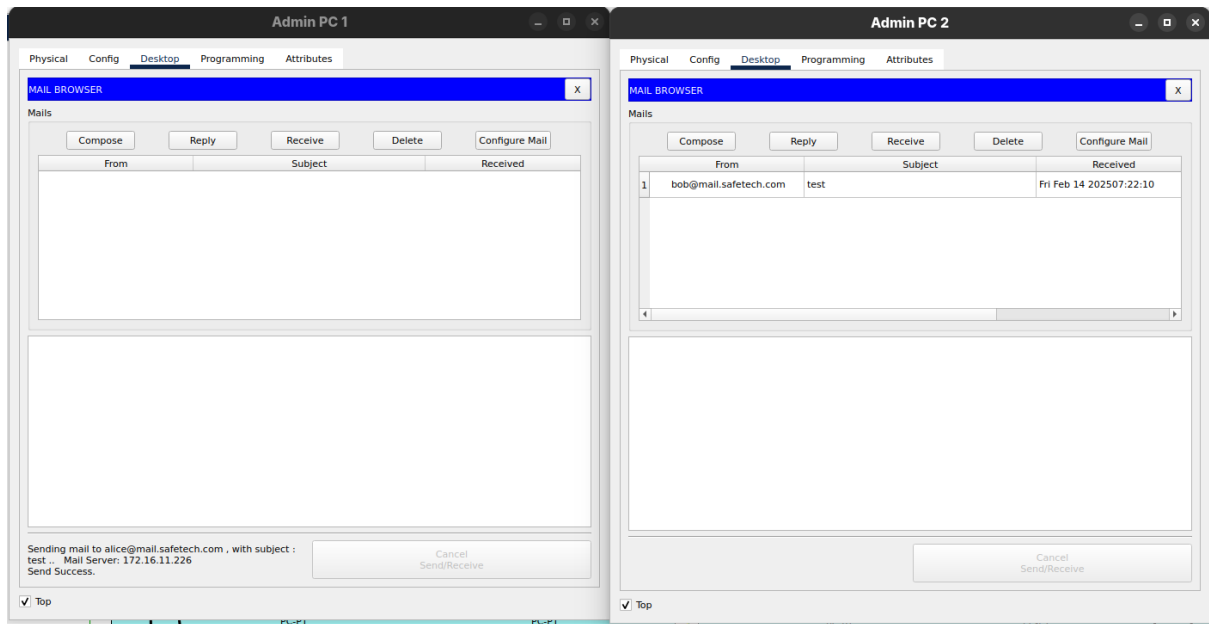


Figure 16: Receiving Email from Bob

Emails have been successfully sent and received, indicating the email server is functioning correctly.

6 Testing and Verification

6.1 Network Connectivity Tests

The network was tested for **inter-VLAN communication**. Below is a **ping test** between the IT and Management departments and a **DHCP request handling**:

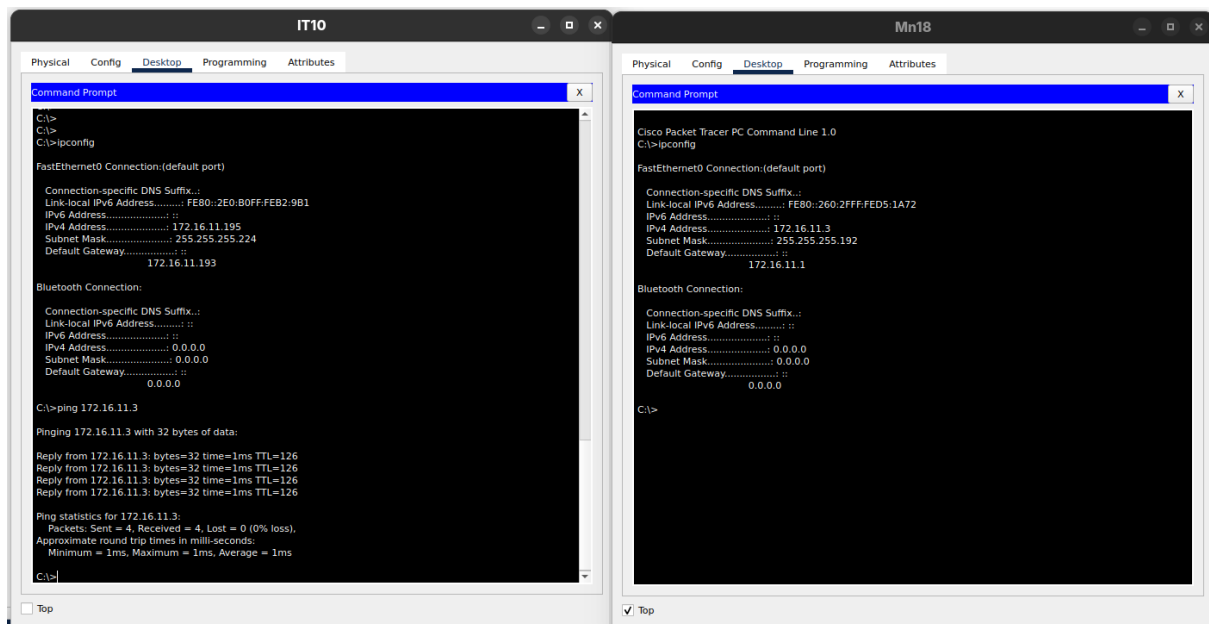


Figure 17: Ping Test: IT to Management

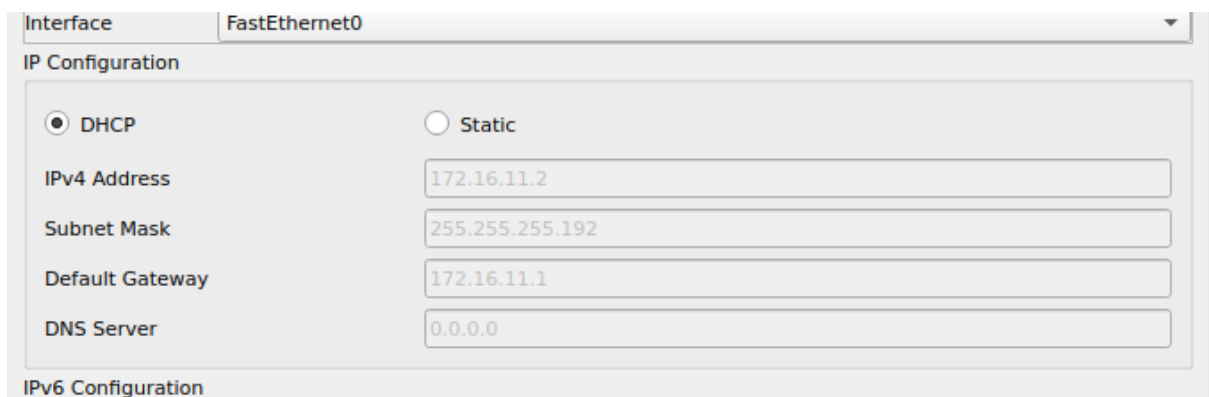


Figure 18: Computer Getting IP Address from DHCP

Anyone can log into any of the routers using SSH:

```
C:\>ssh -l admin 172.16.10.65

Password:
% Login invalid

Password:

+++ ACCESS RESTRICTED +++
Only the faithful may proceed. Unauthorised access will be purged with righteous fire.
BOW BEFORE THE MACHINE GOD! The flesh is weak; the Omnisiahs will is eternal.
+++ GLORY TO THE OMNISIAH +++

Floor_1_Router#
```

Figure 19: Router SSH Login

7 Discussion and Evaluation

7.1 Meeting Requirements

The network design successfully meets the company's requirements. The hierarchical topology with three subnets per floor ensures efficient communication and scalability for future growth. By integrating OSPF and VLANs, the network can support inter-departmental communication without congestion. Additionally, the use of a centralized DHCP server ensures ease of management, while the packet tracer design provides an intuitive visual representation.

7.2 Security Considerations

Security was a key consideration throughout the design process. The implementation of SSH for router access enhances security by ensuring encrypted communication. The use of VLANs isolates traffic between departments, preventing unnecessary exposure. Furthermore, access control lists (ACLs) can be applied to restrict access between VLANs and to the internet, ensuring that only authorized devices and users can access sensitive resources.

7.3 Potential Improvements

While the network design is effective, there are a few areas where improvements can be made:

- Implementing a firewall between departments or at the perimeter for enhanced security.
- Considering the use of redundant links and switches to ensure high availability and fault tolerance.

8 Conclusion

In conclusion, the network design for SAFETech Ltd is robust, scalable, and secure, meeting the current and future needs of the company. The integration of OSPF, VLANs, and DHCP ensures efficient communication and easy management. With some additional security enhancements, such as firewalls and redundant systems, the network can be made even more resilient and secure for long-term use.